

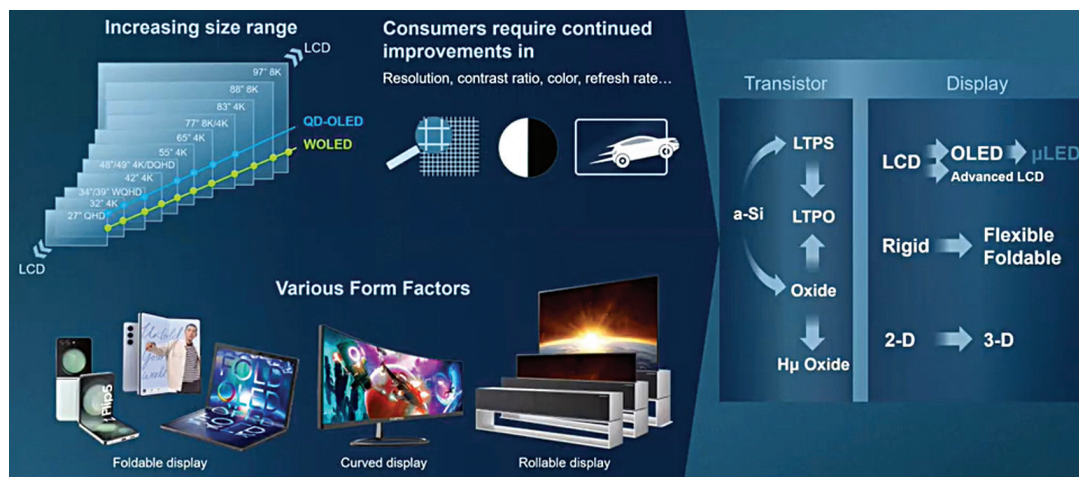


Fig. 2: Display manufacturing technology roadmap

personalised customer experiences, with smart stores engaging consumers via their devices. These innovations are shaping new business models and interactions.

Real-time contaminant detection for food safety

Food safety is another area benefiting from flexible electronics. In India, food adulteration remains a serious issue, and portable, low-cost sensors are making it easier to detect contaminants like pesticides or milk adulteration in real time. This technology empowers consumers and regulatory bodies to enforce quality standards and make informed decisions about food safety.

The potential of flexible electronics extends beyond healthcare:

Wellness tracking. Flexible

sensors embedded in fabrics enable real-time monitoring of vital signs such as heart rate and oxygen levels

Smart textiles. Conductive fabrics can adapt to environmental changes, monitor health, and illuminate under specific conditions

Energy harvesting. Flexible batteries and solar cells power wearables, reducing the need for frequent charging

AR and gaming. Flexible displays enable lightweight, immersive experiences in gaming and augmented reality

In food packaging, tamper-evident tags are enhancing product integrity and security, with trials already underway in India.

As flexible electronics evolve, ensuring affordability and sustainability remains critical. The demand

for eco-friendly manufacturing is driving innovations in biodegradable polymers and energy-efficient production methods, ensuring these technologies remain both cost-effective and environmentally responsible.

Ongoing initiatives in

assistive technologies and rapid diagnostic tools continue to push the limits of what is possible. Wearable medical textiles are particularly promising, integrating fabrics with embedded electronics to monitor health through everyday clothing.

Printable electronics in action, 3D printing and beyond

While most progress has been in 2D printed electronics, 3D printing holds immense potential in this space. Particularly in sensor technology, the ability to print 3D electronics opens up new possibilities. For instance, 3D accelerometers can enhance robotics and automation by enabling more precise, real-time feedback mechanisms.

This evolution extends beyond flexibility—it involves intelligent materials and integration with technologies such as IoT and AI. Flexible, printable electronics are set to become an interactive component of the environment, responding dynamically to users and changing conditions.

The economic impact with India's growing role

The global market for flexible electronics is projected to reach

Challenges and the path forward

While promising, printable electronics face several challenges:

- **Material innovation.** Developing inks with higher conductivity and durability remains a priority
- **Standardisation.** The absence of industry-wide standards complicates integration into existing systems
- **Durability.** Ensuring long-term performance in wearables subjected to daily wear and tear remains a technical challenge

Global efforts are underway to address these challenges, with collaborations between academia, industry, and governments fostering innovation in this domain.